

## Access modifiers

- ❖ Access modifiers can be used to specify the scope or visibility of the class or members of the class.
- ❖ There are three access modifiers:
  - ⇒ Private
  - ⇒ Protected
  - ⇒ Public
- ❖ There are four scope available

| Scope or Visibility | Modifier    |
|---------------------|-------------|
| Private Scope       | Private     |
| Default Scope       | No Modifier |
| Protected           | Protected   |
| Public Scope        | public      |

### Private Scope:

1. When you are using private modifier with the member of the class then the scope of those member will be private scope.
2. Private scope is also called as class scope i.e. Private members must be accessed from the same class where it is declared.
3. Private members cannot be accessed from outside the class, not even from sub class i.e. private members will not be inherited to sub classes.
4. Private modifier can be used only for members of the class not for class itself.

### Default Scope:

1. When you are not using any modifier with the members of the class then the scope of these members will be default Scope.
2. Default scope is also called as packaged scope i.e. default members can be accessed from the same class, subclass and non sub class which are available in same package.
3. Default members cannot be accessed from outside the package i.e. default members will not be inherited to sub class available in outside the package.
4. Default scope is available for both classes and their members.

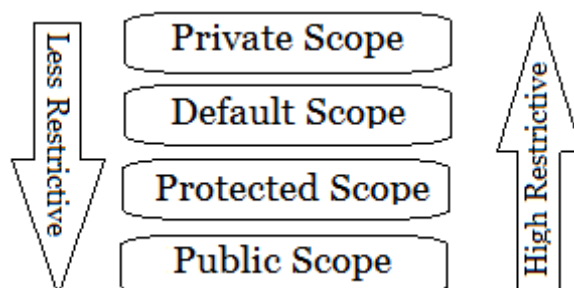
### Protected Scope:

1. When you are using protected modifier with the members of the class then the scope of those members will be protected scope.
2. Protected members can be accessed from the same class, sub class and non sub class which are available in same package.
3. Protected members can be accessed directly or with super or this keyword in sub classes which are available in different package.
4. Protected members cannot be accessed outside the package:
  - ⇒ With object of super class.
  - ⇒ With one sub class object in other sub class.
5. Protected modifier can be used for members of the class not for class itself.
6. Protected member cannot be accessed from the non sub classes which are available in different package.

**Public Scope:**

1. When you are using public modifier with the members of the class then the scope of those members will be public scope.
2. Public members can be accessed from anywhere.
3. Public modifier can be used for both classes and their members.

| Access rights for the different element |                                         |         |         |           |        |
|-----------------------------------------|-----------------------------------------|---------|---------|-----------|--------|
|                                         |                                         | Private | Default | Protected | Public |
| <b>SAME PACKAGE</b>                     | Same class                              | YES     | YES     | YES       | YES    |
|                                         | Subclass                                | NO      | YES     | YES       | YES    |
|                                         | Non subclass                            | NO      | YES     | YES       | YES    |
|                                         | Sub class using super                   | NO      | NO      | YES       | YES    |
| <b>OTHER PACKAGE</b>                    | Sub class using current subclass object | NO      | NO      | YES       | YES    |
|                                         | Sub class using super class object      | NO      | NO      | NO        | YES    |
|                                         | sub class using other sub class object  | NO      | NO      | NO        | YES    |
|                                         | Non sub class another package           | NO      | NO      | NO        | YES    |



**Programs:****A.java**

```

package com.sai.p1;
public class A {
    private int a=10;
    int b=20;
    protected int c=30;
    public int d=40;
    public void showA(){
        System.out.println("A->Show()");
        System.out.println(a);
        System.out.println(b);
        System.out.println(c);
        System.out.println(d);
    }
}

```

**B.java**

```

package com.sai.p1;
public class B extends A{
    public void showB (){
        System.out.println("B->Show()");
        //System.out.println(a);
        System.out.println(b);
        System.out.println(c);
        System.out.println(d);
    }
}

```

**C.java**

```

package com.sai.p1;
public class C {
    public void showC (){
        System.out.println("C->Show()");
        A a1=new A();
        //System.out.println(a1.a);
        System.out.println(a1.b);
        System.out.println(a1.c);
        System.out.println(a1.d);
    }
}

```

**D.java**

```

package com.sai.p2;
package com.sai.p1.A;
public class D extends A{
    public void showD(){
        System.out.println("D->Show()");
        //System.out.println(a);
        //System.out.println(b);
        System.out.println(c);
        System.out.println(d);
    }
}

```

**E.java**

```

package com.sai.p2;
import com.sai.p1.A;
public class E{
    public void showE(){
        System.out.println("E->Show()");
        A a1=new A();
        //System.out.println(a1.a);
        //System.out.println(a1.b);
        //System.out.println(a1.c);
        System.out.println(a1.d);
    }
}

```

**OOPS345.java**

```

package com.sai.p3;
import com.sai.p1.*;
import com.sai.p2.*;
public class OOPS345{
    public static void main(String args[]){
        A ob1=new A();
        ob1.showA();
        B ob2=new B();
        ob2.showB();
        C ob3=new C();
        ob3.showC();
        D ob4=new D();
        ob4.showD();
        E ob5=new E();
        ob5.showE();
    }
}

```

**A1.java**

```

package com.sai.p1;
public class A1{
private int a=10;
int b=20;
protected int c=30;
public int d=40;
public void showA1(){
System.out.println("A1->Show()");
System.out.println("#Direct#");
System.out.println(a);
System.out.println(b);
System.out.println(c);
System.out.println(d);
System.out.println("#A-Object#");
A1 ob1=new A1();
System.out.println(ob1.a);
System.out.println(ob1.b);
System.out.println(ob1.c);
System.out.println(ob1.d);
}
}

```

**B1.java**

```

package com.sai.p1;
public class B1 extends A1{
public void showB1(){
System.out.println("B1->Show()");
System.out.println("#Direct#");
//System.out.println(a);
System.out.println(b);
System.out.println(c);
System.out.println(d);
System.out.println("#A1-Object#");
A1 ob=new A1();
//System.out.println(ob.a);
System.out.println(ob.b);
System.out.println(ob.c);
System.out.println(ob.d);
System.out.println("#B1-Object#");
B1 b1=new B1();
//System.out.println(b1.a);
System.out.println(b1.b);
System.out.println(b1.c);
System.out.println(b1.d);
}
}

```

**C1.java**

```

package com.sai.p1;
public class C1{
public void showC1(){
System.out.println("C1->Show()");
A1 a1=new A1();
System.out.println("#A1-Object#");
//System.out.println(a1.a);
System.out.println(a1.b);
System.out.println(a1.c);
System.out.println(a1.d);
}
}

```

**D1.java**

```

package com.sai.p2;
import com.sai.p1.A1;
public class D1 extends A1{
public void showD1(){
System.out.println("D1->Show()");
System.out.println("#Direct#");
//System.out.println(a);
//System.out.println(b);
System.out.println(c);
System.out.println(d);
}
}

```

```

System.out.println("#D1-Object#");
D1 ob=new D1();
//System.out.println(ob.a);
//System.out.println(ob.b);
System.out.println(ob.c);
System.out.println(ob.d);

System.out.println("#Super keyword #");
//System.out.println(super.a);
//System.out.println(super.b);
System.out.println(super.c);
System.out.println(super.d);

System.out.println("#A1-Object#");
A1 a1=new A1();
//System.out.println(a1.a);
//System.out.println(a1.b);
//System.out.println(a1.c);
System.out.println(a1.d);
}
}

```

**E1.java**

```

package com.sai.p2;
import com.sai.p1.A1;
public class E1
{
    public void showE1()
    {
        System.out.println("E1->Show()");
        A1 a1=new A1();
        //System.out.println(a1.a);
        //System.out.println(a1.b);
        //System.out.println(a1.c);
        System.out.println(a1.d);
    }
}

```

**OOPS346.java**

```

package com.sai.p3;
import com.sai.p1.*;
import com.sai.p2.*;
public class OOPS346{
    public static void main(String args[]){
        A1 ob1=new A1();
        ob1.showA1();
        B1 ob2=new B1();
        ob2.showB1();
        C1 ob3=new C1();
        ob3.showC1();
        D1 ob4=new D1();
        ob4.showD1();
        E1 ob5=new E1();
        ob5.showE1();
    }
}

```

**Hello.java**

```

package com.sai.p3;
public class Java{
    protected int ab=99;
}

```

**OOPS347.java**

```

package kam.sai.p2;
import kam.sai.p1.Hello;
class OOPS347{
    public static void main(String args[]){

```

```

        Hai hai=new Hai();
        hai.show();
    }
}
class Xyz extends Hello{
class Hai extends Hello{
    void show(){
        System.out.println("Hai->Show()");
        Hello h=new Hello();
        System.out.println(h.ab);
        Xyz ref=new Xyz();
        System.out.println(ref.ab);
    }
}

```

**OOPS348.java**

```

class OOPS348{
    public static void main(String args[]){
        A obj=new A();
        obj.x=99;
        System.out.println(obj.x);
    }
}
class A{
    private int x;
}

```

**OOPS349.java**

```

class OOPS349{
    public static void main(String args[]){
        A obj=new A();
        obj.show();
    }
}
class A{
    private int x;
    void show(){
        System.out.println(x);
    }
}

```

**OOPS350.java**

```

class OOPS350{
    public static void main(String args[]){
        A obj=new A(99);
        obj.show();
    }
}

```

```

class A{
    private int x;
    A(int x){
        this.x=x;
    }
    void show(){
        System.out.println(x);
    }
}

```

**OOPS351.java**

```
class OOPS351{
    public static void main(String args[]){
        A obj=new A();
        obj.setx(99);
        System.out.println(obj.getx());
    }
}
class A{
    private int x;
    void setx(int x){
        this.x=x;
    }
    int getx(){
        return this.x;
    }
}
```

**OOPS352.java**

```
class OOPS352{
    public static void main(String args[]){
        B obj=new B(99);
        System.out.println(obj.getx());
    }
}
class A{
    private int x;
    A(int x){
        this.x=x;
    }
    int getx(){
        return this.x;
    }
}
class B extends A{
    B(int x){
        super(x);
    }
}
```

Note: Execute the program and go throw the output.

Help: - First Compile all the class file then compile the program file and then execute the program.

for example to execute program OOPS345.java

Compilation:

```
>javac -d . A1.java
>javac -d . b1.java
>javac -d . B1.java
>javac -d . C1.java
>javac -d . D1.java
>javac -d . E1.java
>javac -d . OOPS345
```

Excution:

```
>java com.sai.p3.OOPS345
```